

Connor Savugot

B.S. in Computer Engineering | North Carolina State University | Graduated May 2026

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Re: Embedded Firmware Engineer

Dear Hiring Manager,

I am applying for the Embedded Firmware Engineer position at your organization. I graduated from North Carolina State University in May 2026 with a B.S. in Computer Engineering and joined AEGIS Power Systems that month as an Embedded Firmware Engineer after completing two summer internships there. I work where low-level firmware, electronics, Embedded Linux, and test hardware meet, and I am comfortable carrying a problem from a datasheet or schematic through implementation, bench diagnosis, and documented handoff.

In my current role, I build and debug TI AM62-based Embedded Linux systems using Yocto/Arago images and toolchains, kernel and device-tree compilation, boot and service diagnostics, systemd services, and external-watchdog supervision. On APU/EPU embedded power subsystems integrating battery hardware, CAN-connected devices, and Linux services, I trace failures across power and wiring, bus traffic, Linux configuration, service logs, and application code. I develop in C/C++, Python, and Bash, use Docker in development workflows, and am improving shared code and tooling so it behaves consistently across Windows and Linux. I also diagnose CAN/CAN FD, IPMI/IPMB, I2C, SPI, and UART issues using logs, register reads, scopes, analyzers, schematics, datasheets, and controlled substitutions.

I held two AEGIS internships: Computer Science Intern in 2024 and Embedded Systems Intern in 2025. Across those roles, I developed TMS320 DSP firmware for transformer-coupled bidirectional buck-boost power stages using twelve PWM A/B pairs, or 24 physical outputs, with independent control of each A signal and match/invert behavior for B. I also developed multiple connected stages of one PIC32/TMS320 product's update workflow: a C++ command-line programmer accepted supplied HEX files, Python/TCP transport handled packetization and checksum validation, and the processors coordinated external SPI-flash and golden-image handling, UART DSP programming, verification, and boot control. In 2024, I built a restricted SSH/chroot interface for TI Yocto/Arago hardware and worked with Python, Bash, RS-232, CAN, and system test hardware.

Separately, my BMS experience comes from my NC State senior design project. I helped define the architecture and requirements, specify interfaces, track risks, and plan validation for a system combining an STMicroelectronics controller, TI BQ-family battery-monitor/ADC, external CAN transceivers, and custom-PCB power, sensing, and telemetry hardware. I evaluated component tradeoffs and coordinated subsystem interfaces, validation, design decisions, and documentation across the team.

Teaching has strengthened the same skills. I supported ECE 306 for three semesters: two in an unofficial teaching support role and one as the official teaching assistant. That separate course used MSP430FR2355 vehicle platforms integrating ADC line sensing, timer/PWM motor control, GPIO/interrupt inputs, Wi-Fi and serial links, LCDs, and custom PCBs. I delivered supplemental lectures, led laboratories, debugged code and hardware, guided projects, and supported students. I also led two semesters of IEEE Open Project Space 2 and separately coordinated and taught through-hole and surface-mount soldering workshops. Guiding students to find their own root causes has made me a more methodical debugger and a clearer engineering teammate.

I would bring hands-on embedded C/C++, Linux, hardware integration, and technical communication experience to your organization. I am especially interested in teams building dependable embedded products where hardware, firmware, and system software must be validated together. I would welcome the opportunity to discuss how I could contribute to your team.

Sincerely,

Connor Savugot